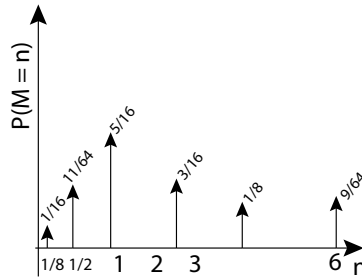


Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026
Homework #3 Solutions

1. (a) The PMF is sketched below. Note that since the value 1 appears twice we need to combine those probabilities. Remember that you should have the values in the right order, you should use arrows (to indicate these are delta functions), and the heights of the arrows should be clearly labeled and proportional.



- (b) We can compute the expected value directly as

$$\begin{aligned}
 E[M] &= \sum_m mP(M=m) \\
 &= 0.125 \cdot \frac{1}{16} + 0.5 \cdot \frac{11}{64} + 1 \cdot \frac{5}{16} + 2 \cdot \frac{3}{16} + 3 \cdot \frac{1}{8} + 6 \cdot \frac{9}{64} \\
 &= \frac{1}{8} \cdot \frac{1}{16} + \frac{1}{2} \cdot \frac{11}{64} + \frac{5}{16} + \frac{6}{16} + \frac{3}{8} + \frac{54}{64} \\
 &= \frac{8 + 88 + 320 + 384 + 384 + 864}{1024} = \frac{2048}{1024} = 2.
 \end{aligned}$$

- (c) We can compute the variance directly as

$$\begin{aligned}
 \text{Var}(M) &= E((M - E(M))^2) \\
 &= (0.125 - 2)^2 \left(\frac{1}{16} \right) + (0.5 - 2)^2 \left(\frac{11}{64} \right) + (1 - 2)^2 \left(\frac{5}{16} \right) \\
 &\quad + (2 - 2)^2 \left(\frac{3}{16} \right) + (3 - 2)^2 \left(\frac{1}{8} \right) + (6 - 2)^2 \left(\frac{9}{64} \right) \\
 &= \frac{3373}{1024} \approx 3.294
 \end{aligned}$$

or equivalently,

- (d)

$$\begin{aligned}
 \text{Var}(M) &= E[M^2] - (E[M])^2, \\
 E[M^2] &= (0.125)^2 \cdot \frac{1}{16} + (0.5)^2 \cdot \frac{11}{64} + 1^2 \cdot \frac{5}{16} + 2^2 \cdot \frac{3}{16} + 3^2 \cdot \frac{1}{8} + 6^2 \cdot \frac{9}{64} \\
 &= \left(\frac{1}{8} \right)^2 \cdot \frac{1}{16} + \left(\frac{1}{2} \right)^2 \cdot \frac{11}{64} + \frac{5}{16} + \frac{12}{16} + \frac{9}{8} + \frac{324}{64} \\
 &= \frac{1}{1024} + \frac{44}{1024} + \frac{320}{1024} + \frac{768}{1024} + \frac{1152}{1024} + \frac{5184}{1024} \\
 &= \frac{7469}{1024}.
 \end{aligned}$$

Since $E[M] = 2$, $(E[M])^2 = 4 = \frac{4096}{1024}$, so

$$\text{Var}(M) = \frac{7469}{1024} - \frac{4096}{1024} = \frac{3373}{1024} \approx 3.294.$$

2. (a) The problem is telling us that the conditional PMF of Z given {old} is binomial, and the conditional PMF of Z given {new} is geometric. Therefore from the properties of these random variables, we know that $E(Z \mid \text{old}) = 5np = 9$ and $E(Z \mid \text{new}) = 8 \cdot \frac{1}{0.4} = 20$. From the first problem we can compute that $P(\text{old}) = \frac{25}{64}$ and $P(\text{new}) = \frac{39}{64}$. Then we can just apply the theorem on total probability to compute that

$$\begin{aligned} E(Z) &= E(Z \mid \text{old}) \cdot P(\text{old}) + E(Z \mid \text{new}) \cdot P(\text{new}) \\ &= 9 \cdot \frac{25}{64} + 20 \cdot \frac{39}{64} \\ &= \frac{1005}{64} \\ &\approx 15.7 \end{aligned}$$

- (b) Computing $E(Z^2)$ directly is a bit of a pain. However, we know that $\text{Var}(Z) = E(Z^2) - (E(Z))^2$, or $E(Z^2) = \text{Var}(Z) + (E(Z))^2$. From the table in the book, we know that $\text{Var}(Z \mid \text{old}) = 5^2 np(1-p) = 31.5$, and $\text{Var}(Z \mid \text{new}) = 8^2 \frac{1-p}{p^2} = 240$. Here we're using the fact that $\text{Var}(aZ) = a^2 \text{Var}(Z)$.

Since we computed the conditional expected values in part (a), we can compute that

$$\begin{aligned} E(Z^2 \mid \text{old}) &= 31.5 + (9)^2 = 112.5 \\ E(Z^2 \mid \text{new}) &= 240 + (20)^2 = 640 \end{aligned}$$

and thus

$$\begin{aligned} E(Z^2) &= E(Z^2 \mid \text{old}) \cdot P(\text{old}) + E(Z^2 \mid \text{new}) \cdot P(\text{new}) \\ &= 112.5 \cdot \frac{25}{64} + 640 \cdot \frac{39}{64} \\ &= 433.9 \end{aligned}$$

3. (a) In a 15-minute exploration, we expect an average of 1.5 red phones, so the random variable has PMF

$$P(N = k) = \frac{1.5^k}{k!} e^{-1.5}, k = 0, 1, 2, \dots$$

We want to compute

$$P(N = 0) = \frac{1.5^0}{0!} e^{-1.5} = e^{-1.5} = 0.223$$

- (b) In a 20-minute exploration, we expect an average of 2 red phones, so the random variable has PMF

$$P(N = k) = \frac{2^k}{k!} e^{-2}, k = 0, 1, 2, \dots$$

We want to compute

$$\begin{aligned}
P(N \geq 3) &= 1 - P(N = 0) - P(N = 1) - P(N = 2) \\
&= 1 - \frac{(2)^0}{0!} e^{-2} - \frac{(2)^1}{1!} e^{-2} - \frac{(2)^2}{2!} e^{-2} \\
&= 1 - 5e^{-2} \\
&= 0.323
\end{aligned}$$

4. (a) The expected value of a uniform distribution is the average of the values it can take on. This may not be one of the original values! So $E[C] = \frac{0+1+2+3+4}{5} = 2$ and $E[L] = \frac{0+1+3+8}{4} = 3$. Then we apply the linearity of expected value to compute

$$\begin{aligned}
E(N) &= E(2C + 3L + 1) \\
&= 2E(C) + 3E(L) + 1 \\
&= 2(2) + 3(3) + 1 \\
&= 14
\end{aligned}$$

- (b) Remember $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$ if X and Y are independent. When the values of a uniform random variable are the integers between a and b , the table tells is the variance is $\frac{1}{12}(L^2 - 1)$, where $L = b - a + 1$. Otherwise, we need to compute the variance directly. A constant value has variance 0!

$$\begin{aligned}
\text{Var}(C) &= \frac{1}{12}(5^2 - 1) \\
&= 2
\end{aligned}$$

$$\begin{aligned}
\text{Var}(L) &= \frac{1}{4}(0^2 + 1^2 + 3^2 + 8^2) - (3)^2 \\
&= 18.5 - 9 \\
&= 9.5
\end{aligned}$$

$$\begin{aligned}
\text{Var}(N) &= \text{Var}(2C + 3L + 1) \\
&= \text{Var}(2C) + \text{Var}(3L) \\
&= 4\text{Var}(C) + 9\text{Var}(L) \\
&= 4(2) + 9(9.5) \\
&= 93.5
\end{aligned}$$